Baichen Du

Last update: October 3, 2023
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Research interests

Topics Phonetics & Phonology: experimental phonology and phonetics, speech production and perception, sound change, cross-modal cue integration.

Education

2023 – 2024 University of California, Berkeley, CA
Undergraduate Visiting Student of Linguistics
Advisor: Dr. Alexandra Pfiffner. *GPA: 4.0 out of 4.0.*

2019 – 2024 University of Hong Kong, HK
Bachelor of Social Sciences in Politics and Public Administration & Linguistics
Advisors: Dr. Jonathan Havenhill, Dr. Youngah Do, Dr. Haohan Chen.
GPA: 3.93 out of 4.3, GPA of Linguistics: 4.12.

Publications & manuscripts

Publications **Du, Baichen.** (To appear). Acoustic distance effect on the perception of sibilants mergers between retroflexes and alveolars in Taiwan Mandarin. In *Proceedings of Seventh Annual Berkeley Undergraduate Linguistics Symposium 2023.* Linguistic Society of University of California, Berkeley, CA

Do, Youngah*, Van Hoey, Thomas, Yu, Xiaoyu, **Du, Baichen**, Tan, Lihui. (In review). Implicit and explicit learning of vowel harmony patterns in phonotactics and alternations

Du, B. (In prep). An audio-visual account for synchronic sound changes towards sibilants merger in Taiwan Mandarin. In *Proceedings of Meetings on Acoustics 184ASA.* Acoustical Society of America, 2023

Thesis **Du, Baichen** (2023). *The backfire effect in the perception and judgement of congruent and incongruent interpretation of political propaganda.* Undergraduate Thesis of Political Psychology

Presentations

2023 **Du, Baichen**, Pfiffner, Alexandra, Johnson, Keith. (2024, Jan). The role of visible articulatory variation in the Mandarin sibilant contrast and merger. Talk to be given at the 98th Annual Meeting of Linguistic Society of America (LSA2024), New York City, NY.

2023 Havenhill, Jonathan, **Du, Baichen**. (2023, Dec). The role of visual cues in sound change: A cross-modality perceptual account for the typological rarity of labial palatalization. Poster to be presented at the 185th Meeting of Acoustical Society of America & 2023 Meeting of the Australian Acoustical Society (ASA185), Sydney, Australia.

2023 **Du, Baichen**, Pfiffner, Alexandra, Johnson, Keith. (2023, Dec). Consonant and vowel rounding: same acoustics, different visuals. Poster to be presented at the 185th Meeting of Acoustical Society of America & 2023 Meeting of the Australian Acoustical Society (ASA185), Sydney, Australia.

2023 **Du, Baichen**, Pfiffner, Alexandra, Johnson, Keith. (2023, Oct). Visible articulatory variation as a cue to sound change: Lip rounding and lip protrusion variability in the Mandarin sibilant merger. Talk to be given at New Ways of Analyzing Variation: Variation in World's Languages (NWAV51), Queens College CUNY, New York City, NY.

2023 **Du, Baichen.** (2023, May). Acoustic distance effect on the perception of sibilants mergers between retroflexes and alveolars in Taiwan Mandarin. Poster presented at the 184th Meeting of Acoustical Society of America 2023 (ASA 184), Chicago, IL. DOI: 10.1121/10.0019189.

2023 **Du, Baichen.** (2023, Apr). An audio-visual account for synchronic sound changes towards sibilants merger in Taiwan Mandarin. Talk given at the Seventh Annual Berkeley Undergraduate Linguistics Symposium 2023, University of California, Berkeley, CA.

2023 **Du, Baichen.** (2023, Apr). The backfire effect in the perception and judgement of congruent and incongruent interpretation of political propaganda. Poster presented at Undergraduate Research Fellowship Program Poster Session: Student Research Posters, University of Hong Kong, HK.

2023 **Du, Baichen** (2023, Sep). Introduction to Advanced R Programming with RMarkdown. The Sociolinguistic Lab of University of California, Berkeley. Slides.

Work-in-progress

In Prep **Du, Baichen**, Pfiffner, Alexandra, Johnson, Keith. Adaptive cue weightings in production and perception of palatal contrast in Korean

Yao, Yao*, **Du, Baichen**, & Johnson, Keith. Production and perception of smiling speech: Implications for sound change.

Havenhill, Jonathan*, **Du, Baichen**. Cross-modal cue integration and phonological evolution: An audio-visual account for palatalization and sound change in diachronic and synchronic Time

Pfiffner, Alexandra*, **Du, Baichen.** The acoustics of plosive lenition in Dutch

Do, Youngah*, Sze, Sui Lung, **Du, Baichen**, Ho, To Tsz, Li, Tak Wang. Bilingual speakers' performance in simultaneous generalization of multiple phonological patterns

Research experience

UC Berkeley

2023 Undergraduate Research Apprenticeship Program — Dr. Alexandra Pfiffner
Supervised Independent Study — Dr. Alexandra Pfiffner

Polytechnic Univ. of HK

2023 Speech and Language Science Lab, — Dr. Yao Yao

Univ. of HK

2023 Dept of Linguistics — Dr. Jonathan Havenhill

2022	Undergraduate Research Fellowship Program	Dr. Haohan Chen
	Dept of Politics and Public Administration	Dr. Haohan Chen
2021/22	Language Development Lab	Dr. Youngah Do & Dr. Jonathan Havenhill.
2021	Undergraduate Research Opportunity Program	Dr. Jonathan Havenhill
	Univ. of Manitoba	
2022	The Baby Language Lab, Univ. of Manitoba	PI: Prof. Melaine Soderstrom.

Grants and fundings

2023	Down Funder Student Travel Funding, Acoustical Society of America. *185th Meeting of Acoustical Society of America & 2023 Meeting of the Australian Acoustical Society, Syndey*	US$1,000
	Instructional Grants at Dept of Linguistics, UC Berkeley. *Audiovisual production and perception of Mandarin sibilants & Cross-modal cue integration and weighting in the perception of Mandarin high front vowel contrast.*	US$600
2022	Transdisciplinary Undergraduate Research Initiative Grant *An audio-visual account for linguistic and social selectivity in the accommodation to sibilant mergers in Mandarin.*	HK$3,830
	Leung Kau Kui Research and Teaching Grant *The backfire effect in the perception and judgement of congruent and incongruent interpretation of political propaganda.*	US$1,300

Honors and Awards

2023	Ms Cecilia Prize of Linguistics	HK$1,000
2022	Globalink Research Award	CA$12,000
	The K.K.Ho Prize	US$2,560
	Dean's Honor List	
2021	Dean's Honor List	
2020	Dean's Honor List	
2019	Soong Ching-ling Foundation Scholarship	US$ 30,700

Skills

Languages English, Mandarin, Cantonese

Programming* R, Praat Scripting, LaTeX, Markdown, HTML, Python

Research Tools RStudio, Praat, PsychoPy, EyeLink, ffmpeg, OpenFace, ELAN, Qualtrics, Montreal Forced Alignment, Emotion and facial expression extraction using CNN, Facial landmarks tracking, TANDAM-STRAIGHT speech synthesis and morphing, Amazon Polly Speech Synthesis, Amazon Mechanical Turk

Research Skills Frequentist and Bayesian statistics (e.g., Linear Mixed Effects Regression, Generalized Mixed Effects Regression), Structural Equation Modeling (e.g., Exploratory and Confirmatory Factor Analysis, Latent Growth Model, Multi-level Modelling), Web Data Extraction, Text Data Processing (e.g., Topic Modelling, Sentiment analysis)

*In a descending order of proficiency

Membership and service

Membership Acoustical Society of America
Linguistic Society of America

Service

Mitacs	Globalink Research Ambassador
Undergraduate Buddy Program	Mentor to Fu Jiani
Chinese Culture Committee	Chariperson
Social Service Fund	External Secretary
Pad for Hope Foundation	Teaching Assistant
University Service Centre of CUHK	Helper

Key courses

Linguistics Phonetics (A), Advanced Phonetics (A), Experimental Phonetics (A+), Phonology (A+), Phonetic-Phonology interface (A+), Variation analysis (A-), Advanced Sociolinguistics (A), Laboratory linguistics (A), Second language research (A+), Languages of the world (A+), Grammatical description (A+), Semantics (A-), Language Revitalization (Pending).

Methodology Research Methods in Social Sciences (A+), Data Science in Politics and Public Administration (A+), Data Analysis with R (A), Policy and Program Evaluation (A-), Advanced Methods in Psychological Science (Pending)